Varshith Ganta

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.56
Sri Chaitanya Jr Kalasala	Class XII (Telangana State Board of Intermediate Education)	2023	97.20%
Sri Chaitanya Techno school	Class X (Board of Secondary Education, Telangana)	2021	10.00%

WORK EXPERIENCE

Quantum Engineer Intern at **Accelequant India Pvt. ltd.**

- Built a unified **benchmarking pipeline** that ingests optimization problem instances across multiple formats (.mps, .mps.gz, .mps.zip, .lp, raw QUBO) from local storage or an S3-hosted database, normalizing all inputs into a standardized representation consumed uniformly by multiple solver backends.
- Extended the benchmarking pipeline so two independent optimization solvers could be swapped in through the same interface without any solver specific changes elsewhere in the pipeline, and built **automatic result aggregation** with merge logic plus computed cross solver comparison metrics (Hamming distance, best-solution-found time).
- **Obfuscated** different pipelines which were being used and developed an **automated obfuscation tool** that recursively collects Python modules across a multi-folder codebase, generates build configuration, and compiles them via **Cython**, replacing a manual per module obfuscation workflow used to protect the codebase pre-deployment. Evaluated obfuscating tools for integration effort and functional preservation.

PROJECTS

Document Retrieval & Search Engine

- Built a command-line document search engine that ingests PDF/DOCX/TXT files and returns ranked results for a query, implementing the full classical IR pipeline (tokenization ,normalization , stop-word filtering ,inverted indexing) without using any existing search library.
- Implemented TF-IDF and BM25 ranking algorithms from scratch, including IDF weighting, term-frequency saturation, and document-length normalization. Benchmarked both methods against each other on the same queries.
- Designed a JSON-based index cache so documents are processed once and reused across queries, avoiding redundant re-indexing on every run.

Used Car Price Prediction using Neural Networks

Designed and implemented a deep learning model to estimate the resale value of used cars based on factors like age, mileage, fuel type, and transmission. Focused on feature engineering and neural network tuning to build a robust, real-world price prediction system.

- Built a multi-layer neural network using Keras with ReLU activations and dropout regularization to prevent overfitting.
- Engineered meaningful input features, applied one-hot encoding and standardization for optimal model input.
- Tuned hyperparameters and trained the model using the Adam optimizer; evaluated performance using RMSE to measure predictive accuracy.

Crop Yield Analysis

Explored and transformed agricultural yield data to prepare it for predictive modeling. Focused on understanding data distribution, handling skewness, and identifying key features affecting crop output.

- Analyzed dataset characteristics using pandas, matplotlib, and seaborn to explore crop yield patterns.
- Applied logarithmic transformation to stabilize variance and normalize the target variable.
- Visualized distributions and feature relationships to identify potential predictors for future modeling.

SKILLS

Languages: C, C++, Python, SQL

Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems

Tools & Technologies: Git, Linux, VS Code, Jupyter Notebook, LaTeX

Machine Learning: Neural Networks (Keras/TensorFlow), Regression, Decision Trees, Random Forests

Other: Qiskit, Cython, PyArmor

POSITIONS OF RESPONSIBILITY

Public Relations Coordinator- IR Cell

- Managed social media content and engagement for international collaborations.

- Designed, posted, and tracked digital campaigns for events.

Core Member-Aero

- Part of a hands-on team building an RC aircraft using constrained materials.

- Attended and supported internal workshops on flight theory and structural design.

Core Member- Cepheid

- Helped organize stargazing and outreach sessions for campus members.

- Contributed to event planning, logistics, and offline promotions.

EXTRACURRICULAR

- **NSS Member :** Participated in social outreach initiatives such as cleanliness drives, awareness campaigns, and community service programs organized by the college NSS unit.

- **eSports**: Won Silver Medal in college Call of Duty: Mobile tournament and also took part in BGMI events.

- **Cricket and Badminton**: Regularly playing both sports on campus; enjoying team games and staying active

RELEVANT COURSES

Computer Science: Introduction to Programming

Mathematics: Numerical Methods, Applied Mathematics, Calculus, Elementary Linear Algebra

Online Courses:

Machine Learning: 1) Supervised Machine Learning **2)** Advanced Learning algorithms **3)** Unsupervised Machine Learning